Folded Transport MCMC: Eliminating Label Switching by Sampling on a Fundamental Domain

Abstract

In Bayesian mixture models and other exchangeable-component models, the posterior is invariant under permutation of component labels, creating $m!$ equivalent modes—a phenomenon known as label switching. Standard MCMC methods either mix poorly across these modes or rely on post-hoc relabelling that cannot guarantee the sampler has converged. We propose *Folded Transport MCMC* (FolT-MCMC), which eliminates label switching before sampling by restricting the Markov chain to a *fundamental domain*—a sorted or reflected subspace containing exactly one representative from each symmetric mode. The proposal distribution is a learned normalising flow whose density is symmetrised over the group orbits, ensuring correct targeting on the reduced space. We show that this construction preserves a computable convergence diagnostic based on the oscillation of the log-density ratio, and that this diagnostic becomes sharper on the fundamental domain whenever the original-space flow under-covers one or more symmetric modes. Experiments on Gaussian mixtures ($d = 2$–20), label-switching targets (up to 24 equivalent modes), a standard Bayesian three-component mixture posterior, and real accelerometer data from a supertall building demonstrate improvement ratios of $2\times$ to $145\times$ in the convergence diagnostic, with the folded diagnostic remaining stable across dimensions while the unfolded diagnostic collapses.

Keywords: Markov chain Monte Carlo; label switching; mixture models; normalising flows; convergence diagnostic; Bayesian structural identification.

1 Introduction

1.1 The label-switching problem

Bayesian mixture models with m exchangeable components define posterior distributions with $m!$ equivalent modes: for every posterior sample $(\mu_1, \sigma_1, \ldots, \mu_m, \sigma_m)$, every permutation of the component indices $(\mu_{\tau(1)}, \sigma_{\tau(1)}, \ldots)$ has exactly the same posterior density. This phenomenon, known as *label switching* [Stephens, 2000, Jasra et al., 2005, Celeux et al., 2000], is a central challenge in Bayesian computation for mixture models, hidden Markov models, factor models, and structural identification problems with closely-spaced modes [Au, 2011, Yang et al., 2019].

The standard response is *post-hoc relabelling*: run MCMC on the full $m!$-modal posterior, then reorder the samples after the fact using a loss function [Stephens, 2000], an equivalence-class algorithm [Papastamoulis and Iliopoulos, 2010], or random permutations at each step [Frühwirth-Schnatter, 2001]. These methods improve the interpretability of the output but do not address the fundamental difficulty: the Markov chain must still traverse $m!$ symmetric modes, and there is no guarantee that it has done so adequately.

1.2 Our approach: sampling on a fundamental domain

A cleaner alternative is to sample directly on a *fundamental domain*—a subspace that contains exactly one representative from each group of equivalent modes. For a three-component mixture, sorting the components by their first parameter ($\mu_1 \leq \mu_2 \leq \mu_3$) maps the $3! = 6$ equivalent modes to a single sorted mode on the "sorted chamber." The posterior restricted to this domain, which we call the *folded target*, integrates to one and retains all the inferential content of the original posterior without the redundant copies.

This idea has appeared informally in the mixture literature [Celeux et al., 2000, Frühwirth-Schnatter, 2001], but has not previously been combined with learned proposal distributions that can efficiently explore the folded target. We introduce *Folded Transport MCMC* (FolT-MCMC), which pairs the fundamental-domain construction with a *normalising flow*—a learnable, invertible transformation that maps a simple distribution (e.g., a standard normal) to an approximation of the target. Normalising flows have been used as proposal distributions for Metropolis–Hastings samplers [Parno and Marzouk, 2018, Hoffman et al., 2019, Gabrié et al., 2022]; FolT-MCMC trains the flow on the folded target and symmetrises its density over the group orbits to form a *quotient proposal* that covers the fundamental domain.

1.3 A computable convergence diagnostic

A distinctive feature of FolT-MCMC is that the folded sampler comes with a computable *convergence diagnostic* based on the oscillation of the log-density ratio between target and proposal.

For an independence Metropolis–Hastings sampler with target density π and proposal density q, the spectral gap γ (which controls the geometric convergence rate) is bounded below by

$$\gamma \;\geq\; \frac{2}{1+\exp(\operatorname{osc}(h))}, \qquad h = \log \pi - \log q,$$

where $\operatorname{osc}(h) = \sup h - \inf h$ is the oscillation of the log-density ratio [Mengersen and Tweedie, 1996]. When the flow perfectly matches the target, the ratio is constant and $\gamma = 1$; as the approximation degrades, the oscillation grows and γ shrinks exponentially.

Computing this bound from finite samples requires controlling the oscillation over a high-probability region of the posterior (a credible set). Recent work has developed a framework for doing so: a coverage-based empirical oscillation bound combined with Lipschitz interpolation yields a computable upper bound on $\operatorname{osc}(h)$ from posterior samples alone [Hu, 2026b,a]. We call the resulting quantity a *convergence diagnostic* rather than a spectral-gap certificate, because it controls the oscillation on the high-probability core of the posterior but not on the tails (see Section 4 for a precise statement).

The key observation motivating this paper is that *label switching destroys this diagnostic*. When the posterior has $m!$ symmetric modes and the flow can only approximate one of them, the log-density ratio is large on the off-modes, inflating the oscillation and rendering the diagnostic vacuous (i.e., the bound gives $\gamma \approx 0$ even when the sampler is working well empirically). By folding onto the fundamental domain, FolT-MCMC removes the cross-mode oscillation, restoring a meaningful diagnostic.

1.4 Contributions

The three main contributions are:

1. **Fundamental-domain sampler.** For a posterior π invariant under a finite group G (e.g., $G = S_m$ for m-component mixtures), we define the folded target on the fundamental domain, prove that its credible sets are simply the intersection of the original credible sets with the domain, and construct a symmetrised proposal from a learned normalising flow (Section 3).
2. **Sharper convergence diagnostic.** We show that the oscillation-based diagnostic transfers to the fundamental domain with improved inputs: the credible-set diameter does not increase, the density floor improves by a factor of $|G|$, and the covering radius decreases. When the flow trained on the original space under-covers one or more symmetric modes (as observed in our experiments), the folded diagnostic is sharper (Section 4).
3. **Empirical validation.** On Gaussian mixtures ($d = 2$–20), label-switching targets (up to 24 equivalent modes), a standard Bayesian three-component mixture posterior, and real accelerometer data from a supertall building during Typhoon Mangkhut, the diagnostic improvement ratio ranges from $2\times$ to $145\times$. The folded diagnostic is empirically stable across dimensions, while the unfolded diagnostic collapses. On the Bayesian mixture benchmark, post-hoc relabelling recovers correct point estimates but provides no convergence guarantee, demonstrating that quotient-space sampling is needed for diagnosing posterior convergence (Sections 5–6).

1.5 Outline

Section 2 provides background on normalising flows as MCMC proposals, the oscillation-based convergence diagnostic, and symmetry in Bayesian models. Section 3 develops the FolT-MCMC framework. Section 4 establishes the theoretical guarantees. Section 5 presents the experiments. Section 6 applies FolT-MCMC to structural modal identification with real typhoon data. Section 7 discusses limitations and extensions.

Terminology and notation. Throughout, we use the following terms: *normalising flow*—a learnable invertible map $T\colon \mathbb{R}^d \to \mathbb{R}^d$ that transforms a simple base distribution to an approximation of the target; *independence Metropolis–Hastings (IMH)*—a Metropolis–Hastings sampler whose proposals are drawn independently of the current state; *spectral gap* (γ)—a number in $[0, 1]$ measuring the geometric convergence rate of a Markov chain; *oscillation* ($\mathrm{osc}(h)$)—the range $\sup h - \inf h$ of the log-density ratio $h = \log \pi - \log q$; *fundamental domain* (D)—a subspace containing one representative per symmetric mode (e.g., the sorted chamber); *folded target* (π_F)—the posterior restricted and renormalised to the fundamental domain; *convergence diagnostic*—the computable lower bound on γ obtained from finite samples via the oscillation framework. All acronyms are defined at first use.

2 Background and notation

2.1 Transport MCMC and independence Metropolis–Hastings

A normalising flow $T\colon \mathbb{R}^d \to \mathbb{R}^d$ is a learnable diffeomorphism that pushes a simple base density q_0 (typically $\mathcal{N}(0, I_d)$) to an approximation $q = T_\# q_0$ of a target density π. In transport MCMC, the flow serves as the proposal mechanism for a Markov chain whose stationary distribution is exactly π, correcting any approximation error in q through the Metropolis–Hastings acceptance step [Parno and Marzouk, 2018, Hoffman et al., 2019, Gabrié et al., 2022].

The simplest such chain is *independence Metropolis–Hastings* (IMH): given the current state θ, propose $\theta' \sim q$ and accept with probability $\min(1, e^{h(\theta)-h(\theta')})$, where $h = \log \pi - \log q$ is the log-density ratio. IMH is particularly well suited to transport proposals because the proposal is cheap to evaluate (a single forward pass through the flow) and the acceptance probability depends only on the quality of the density-ratio fit.

The spectral gap γ of the IMH chain—measuring the rate of geometric convergence—is bounded below by the Mengersen–Tweedie inequality [Mengersen and Tweedie, 1996]:

$$\gamma \;\geq\; \frac{2}{1+\exp\bigl(\operatorname{osc}(h)\bigr)}, \qquad \operatorname{osc}(h) := \sup_{\theta} h(\theta) - \inf_{\theta} h(\theta). \tag{1}$$

When the flow approximation is perfect ($q = \pi$), $\operatorname{osc}(h) = 0$ and $\gamma = 1$; as the approximation degrades, $\operatorname{osc}(h)$ grows and γ shrinks exponentially.

2.2 A computable convergence diagnostic

The oscillation $\operatorname{osc}(h)$ is not directly computable from finite samples, because both the supremum and infimum range over the entire support of π. Recent work [Hu, 2026b] provides a computable upper bound on the oscillation over a highest posterior density (HPD) credible set $\mathcal{K}_\alpha$—the smallest region containing $(1-\alpha)$ of the posterior mass—yielding a computable core lower-bound diagnostic, using three ingredients.

Spectrally normalised RealNVP. The flow T is a RealNVP—a coupling-based normalising flow architecture [Dinh et al., 2017] in which each layer transforms half of the coordinates conditionally on the other half—built from L coupling layers. Each conditioner network is *spectrally normalised*: a weight-matrix constraint that bounds the Lipschitz constant of the network [Miyato et al., 2018], here combined with a scale clip c. This gives a per-layer bi-Lipschitz bound, ensuring that the Jacobian and its log-determinant are well controlled.

Empirical oscillation theorem. Given n i.i.d. samples $\{Z_i\}$ from π and the $(1-\alpha)$-HPD set $\mathcal{K}_\alpha$, the empirical oscillation theorem of Hu [2026b] states that with probability $\geq 1-\delta$:

$$\operatorname{osc}_{\mathcal{K}_\alpha}(h) \;\leq\; \underbrace{\max_i h(Z_i) - \min_i h(Z_i)}_{\widehat{\operatorname{osc}}_n} \;+\; \underbrace{2\,M\,\varepsilon^*}_{\text{interpolation error}}, \tag{2}$$

where $M = \sup_{\mathcal{K}_\alpha} \|\nabla h\|$ is the local gradient constant and ε^* is the minimal covering radius satisfying a covering condition that depends on $\operatorname{diam}(\mathcal{K}_\alpha)$, $\pi_{\min}$, and n. The righthand side is fully computable from samples and network parameters.

Quantile-core diagnostic. The HPD-core oscillation bound (2) can still be dominated by extreme density-ratio values among the retained core samples. The quantile-core extension of Hu [2026a] replaces the full HPD set with a ρ-trimmed core that excludes the highest-oscillation fraction, yielding a tighter *quantile-core* (QC) diagnostic. In all experiments of the present paper, we report the quantile-core diagnostic at $\rho = 0.05$.

Limitation: multimodality. When π has k well-separated modes and the flow can only Gaussianise one of them, the log-density ratio h is large on the off-modes (where $q \ll \pi$), inflating $\widehat{\text{osc}}_n$ by a cross-mode gap that grows with mode separation and dimension. This is the fundamental barrier that FolT-MCMC is designed to overcome.

2.3 Symmetry in Bayesian inference

Many Bayesian posteriors possess a finite symmetry group G under which the likelihood and prior are jointly invariant. The resulting symmetric multimodality is a well-known obstacle to MCMC mixing and inference.

Label switching. In m-component mixture models, the likelihood is invariant under permutation of component labels ($G = S_m$, $|G| = m!$). This creates $m!$ equivalent posterior modes, between which standard MCMC chains mix poorly [Celeux et al., 2000, Jasra et al., 2005]. Existing solutions are predominantly *post-hoc*: after running MCMC in the full $m!$-modal space, the samples are relabelled using a loss function [Stephens, 2000], an equivalence-class representative algorithm [Papastamoulis and Iliopoulos, 2010], or random permutation at each step [Frühwirth-Schnatter, 2001]. These methods do not address the poor mixing of the underlying chain, nor do they provide a convergence diagnostic.

Structural modal identification. In Bayesian operational modal analysis of buildings and bridges, closely-spaced structural modes create approximate label-switching symmetry: the posterior over m sets of modal parameters (frequency, damping ratio) is nearly

S_m-invariant when the modes have similar frequencies [Au, 2011, Yang et al., 2019]. This is a key motivation for the present work.

Equivariant normalising flows. An alternative approach constructs flows that respect the symmetry by design: $T(g \cdot z) = g \cdot T(z)$ [Köhler et al., 2020, Rezende et al., 2019]. Such equivariant flows reduce parameter redundancy but must still represent all $|G|$ modes. FolT-MCMC takes the complementary approach of *eliminating* the symmetric modes before sampling, reducing the target to a single representative per orbit. The two strategies are orthogonal and could in principle be combined.

3 The FolT-MCMC framework

Let π be a target density on $\mathbb{R}^d$ that is invariant under a finite group G acting on $\mathbb{R}^d$: $\pi(g \cdot \theta) = \pi(\theta)$ for every $g \in G$ and almost every θ. Write $s = |G|$. Typical examples are label-switching symmetry in mixture models ($G = S_m$, the symmetric group on m components, $s = m!$) and reflection symmetry ($G = \mathbb{Z}_2$, $s = 2$).

The central idea of FolT-MCMC is to *eliminate* the symmetric multimodality of π before sampling. We restrict the target to a fundamental domain, obtaining the folded target introduced in Section 1. We then run an independence Metropolis–Hastings sampler on this domain, using a learned normalising flow as proposal, and compute the convergence diagnostic from the oscillation of the log-density ratio.

3.1 The folded target

A *fundamental domain* for G is a measurable set $D \subset \mathbb{R}^d$ such that $\mathbb{R}^d = \bigcup_{g \in G} g \cdot D$ with pairwise disjoint interiors and $\pi(g \cdot D \cap g' \cdot D) = 0$ for $g \neq g'$.

Definition 3.1 (Folded target)**.** The *folded* or *quotient target* on D is

$$\pi_F(z) \;=\; s \cdot \pi(z), \qquad z \in D. \tag{3}$$

The corresponding potential is $U_F(z) = -\log \pi_F(z) = U(z) - \log s$.

Normalisation is immediate: by the fundamental-domain decomposition, $\int_D \pi_F = s \int_D \pi = s \cdot (1/s) = 1$. The density π_F retains one representative from each G-orbit of modes, so the symmetric multimodality induced by the group action is absent; non-symmetric multimodality, if present, may remain.

The next result shows that folding preserves the highest posterior density (HPD) structure exactly.

Proposition 3.2 (HPD identity)**.** *Let u_α be the $(1-\alpha)$-quantile of $U(\Theta)$ under $\Theta \sim \pi$, defining the HPD set $\mathcal{K}_\alpha = \{\theta \in \mathbb{R}^d : U(\theta) \le u_\alpha\}$. Then:*

1. $u_\alpha^F = u_\alpha - \log s$;
2. $\mathcal{K}_\alpha^F := \{z \in D : U_F(z) \le u_\alpha^F\} = \mathcal{K}_\alpha \cap D$ *(a.e.).*

Proof. For any Borel function $\varphi\colon \mathbb{R} \to \mathbb{R}$, the G-invariance of π and the fundamental-domain decomposition give

$$\int_{\mathbb{R}^d} \varphi(U(\theta))\, \pi(\theta)\, \mathrm{d}\theta = \sum_{g \in G} \int_{gD} \varphi(U(\theta))\, \pi(\theta)\, \mathrm{d}\theta = s \int_D \varphi(U(z))\, \pi(z)\, \mathrm{d}z = \int_D \varphi(U(z))\, \pi_F(z)\, \mathrm{d}z.$$

Hence $U(Z_F)$ under $Z_F \sim \pi_F$ has the same distribution as $U(\Theta)$ under $\Theta \sim \pi$, so the two quantiles coincide: the $(1-\alpha)$-quantile of $U_F(Z_F) = U(Z_F) - \log s$ is $u_\alpha - \log s$. Part (ii) follows directly: $\{U_F \le u_\alpha^F\} = \{U - \log s \le u_\alpha - \log s\} = \{U \le u_\alpha\}$. □

Corollary 3.3. *(a)* $\mathrm{diam}(\mathcal{K}_\alpha^F) \le \mathrm{diam}(\mathcal{K}_\alpha)$. *(b) The density floor satisfies* $\pi_{\min}^F := \inf_{\mathcal{K}_\alpha^F} \pi_F = s \cdot \inf_{\mathcal{K}_\alpha \cap D} \pi \ge s \cdot \pi_{\min}$*, where* $\pi_{\min} := \inf_{\mathcal{K}_\alpha} \pi$.

Using essential infima with respect to the Lebesgue measure, the G-invariance of π ensures $\operatorname{ess\,inf}_{\mathcal{K}_\alpha \cap D} \pi = \operatorname{ess\,inf}_{\mathcal{K}_\alpha} \pi$, so $\pi_{\min}^F = s\, \pi_{\min}$. With ordinary infima this equality holds for closed fundamental domains; in general we use the conservative bound $\pi_{\min}^F \ge s\, \pi_{\min}$. For label switching with $m = 4$ components, $s = 24$ and the floor is 24 times higher.

3.2 Quotient proposal

Let $T_F\colon \mathbb{R}^d \to \mathbb{R}^d$ be a normalising flow (e.g. spectrally normalised RealNVP) trained on folded samples projected onto D, and let $q_{T_F} = (T_F)_\# \mathcal{N}(0, I_d)$ denote the induced density on $\mathbb{R}^d$.

Definition 3.4 (Folded (quotient) proposal)**.** The *quotient proposal* on D is

$$q_F(z) \;=\; \sum_{g \in G} q_{T_F}(g \cdot z), \qquad z \in D. \tag{4}$$

Normalisation follows from the fundamental-domain decomposition: $\int_D q_F = \sum_g \int_D q_{T_F}(g \cdot z)\,\mathrm{d}z = \sum_g \int_{gD} q_{T_F}(\theta)\,\mathrm{d}\theta = \int_{\mathbb{R}^d} q_{T_F} = 1$. Intuitively, q_F is the unique G-symmetric density on $\mathbb{R}^d$ obtained by summing q_{T_F} over the group orbits; restricted to D, it gives the proposal used by the sampler.

The log-density ratio governing the Metropolis–Hastings acceptance is

$$h_F(z) \;=\; \log \frac{\pi_F(z)}{q_F(z)} \;=\; \log\bigl(s \cdot \pi(z)\bigr) \;-\; \log\Bigl(\sum_{g \in G} q_{T_F}(g \cdot z)\Bigr). \tag{5}$$

Remark 3.5 (Relation to unfolded MH). Independence MH on D with target π_F and proposal q_F is a valid Markov chain with stationary distribution π_F. It is *not* identical to running MH on $\mathbb{R}^d$ and projecting samples onto D, because $\pi_F(z)/q_F(z)$ generally differs from $\pi(z)/q_{T_F}(z)$. The two coincide when q_{T_F} is itself G-invariant. In practice, when q_{T_F} concentrates on a single mode, the off-mode terms $q_{T_F}(g \cdot z)$ for $g \neq e$ are negligible for z deep in D, and the two ratios agree approximately.

3.3 Algorithm

FolT-MCMC combines folding, transport, sampling, and diagnosis in the workflow below.

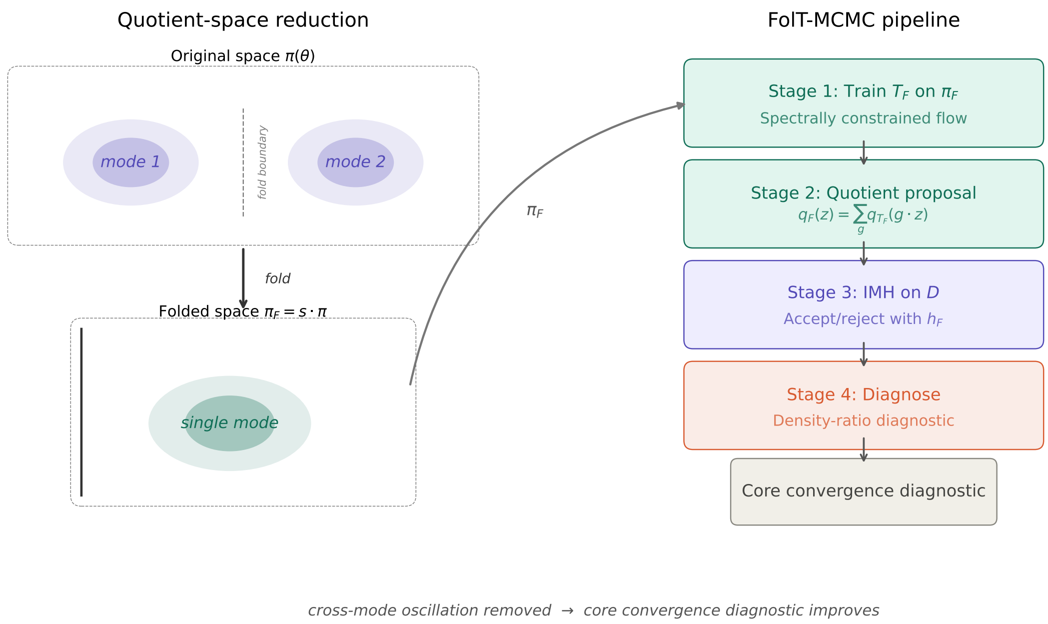

Figure 1: FolT-MCMC pipeline. Left: the original posterior π has s symmetric modes; folding onto the fundamental domain D produces the quotient target π_F with a single representative mode. Right: the four-stage pipeline trains a transport map on π_F, forms the quotient proposal, runs independence MH on D, and computes the convergence diagnostic from the oscillation framework.

Algorithm: FolT-MCMC

Input: target π with known symmetry group G, fundamental domain D.

Stage 1 (Fold and train).
Draw reference samples $\{z_i\}_{i=1}^{N}$ from π_F (fold exact or MCMC samples into D).
Train a spectrally normalised RealNVP flow T_F on π_F using negative log-likelihood (NLL) + oscillation regularisation.
Form the quotient proposal q_F via (4).

Stage 2 (Sample).
Run independence Metropolis–Hastings on D:
Propose $z' \sim q_F$ by sampling $z_0 \sim \mathcal{N}(0, I)$, computing $\theta = T_F(z_0)$, setting $z' = \mathrm{proj}_D(\theta)$.
Accept with probability $\min\big(1,\, e^{h_F(z_{\mathrm{curr}}) - h_F(z')}\big)$.

Stage 3 (Diagnose).
Compute the empirical oscillation $\widehat{\mathrm{osc}}_n^F$ and gradient supremum $\widehat{M}_n^F$ from an independent diagnostic set on D.
Solve the covering condition (9) for ε_F^*.
Report the quantile-core convergence diagnostic $\mathrm{QC}\,\underline{\gamma}_F \geq 2/(1 + \exp(\widehat{\mathrm{osc}}_n^F + 2\widehat{M}_n^F \varepsilon_F^*))$.

Output: posterior samples on D; quantile-core diagnostic $\mathrm{QC}\,\underline{\gamma}_F$.

We highlight two concrete instantiations.

Reflection fold ($G = \mathbb{Z}_2$). The non-identity element acts by negating the first coordinate: $g \cdot \theta = (-\theta_1, \theta_2, \ldots, \theta_d)$. The fundamental domain is $D = \{\theta_1 \geq 0\}$ and $\mathrm{proj}_D(\theta) = (|\theta_1|, \theta_2, \ldots, \theta_d)$. The quotient proposal reduces to $q_F(z) = q_{T_F}(z) + q_{T_F}(-z_1, z_2, \ldots, z_d)$.

Permutation fold ($G = S_m$). Each $\theta \in \mathbb{R}^d$ is partitioned into m blocks of p parameters ($d = mp$), and $G = S_m$ permutes blocks. The fundamental domain is the sorted chamber $D = \{\theta : \theta_1^{(1)} \leq \theta_1^{(2)} \leq \cdots \leq \theta_1^{(m)}\}$, where $\theta_1^{(j)}$ is the first parameter (e.g. natural frequency) of the jth block. Projection sorts the blocks by this parameter.

3.4 Quotient metric and covering

The empirical oscillation framework of Hu [2026b] relies on a covering lemma (Lemma 5.2 therein) that lower-bounds the probability mass of balls intersected with the HPD set. On the fundamental domain D, Euclidean balls near the fold boundary ∂D may be truncated, potentially losing mass. We resolve this by working in the *quotient metric*.

Definition 3.6 (Quotient metric)**.** For $\theta, \theta' \in \mathbb{R}^d$, define

$$d_G(\theta, \theta') \;=\; \min_{g \in G} \|\theta - g \cdot \theta'\|. \tag{6}$$

This descends to a well-defined metric on the orbit space $\mathbb{R}^d/G$.

Since $d_G \leq \|\cdot\|$ (take $g = e$), any Euclidean ε-cover is automatically a quotient ε-cover, giving the covering-number bound

$$\mathcal{N}_G(\mathcal{K}_\alpha^F, \varepsilon) \;\leq\; \mathcal{N}_E(\mathcal{K}_\alpha^F, \varepsilon) \;\leq\; \Big(\frac{2\mathrm{diam}(\mathcal{K}_\alpha^F)}{\varepsilon} + 1\Big)^d. \tag{7}$$

The advantage of the quotient metric emerges when bounding the *ball mass*. For a point $z \in D$ whose stabiliser is trivial ($\mathrm{Stab}(z) = \{e\}$), the quotient ball $B_G(z, \varepsilon) := \{z' : d_G(z, z') \leq \varepsilon\}$ corresponds in $\mathbb{R}^d$ to the union $\bigcup_{g \in G}\big(B(g \cdot z, \varepsilon) \cap g \cdot D\big)$, which has total volume equal to that of a full Euclidean ball $V_d\, \varepsilon^d$—no half-ball or boundary correction is needed. For well-separated mixture posteriors, the HPD set $\mathcal{K}_\alpha^F$ lies in the interior of D and all stabilisers are trivial; we formalise this as a regularity condition.

Assumption 3.7 (Generic stabiliser)**.** $|\mathrm{Stab}(z)| = 1$ *for every* $z \in \mathcal{K}_\alpha^F$*.*

Assumption 3.7 holds whenever the mode centres are distinct after projection into D and the modes are sufficiently well separated that the HPD set does not touch the fold boundary. When it fails (e.g. at fixed-point strata of the group action), the ball-mass bound acquires a correction factor $1/h_{\max}$ where $h_{\max} = \max_{z\in\mathcal{K}_\alpha^F} |\mathrm{Stab}(z)|$; see Supplement C.3.

The remaining ingredient is a Lipschitz property of h_F under the quotient metric.

Lemma 3.8 (Quotient Lipschitz constant)**.** *Let h_F be extended to $\mathbb{R}^d$ by $h_F(\theta) = \log\big(s\pi(\theta)\big) - \log\big(\sum_g q_{T_F}(g\theta)\big)$, and let $M_F = \sup\big\{\|\nabla h_F(w)\| : d_G(w, \mathcal{K}_\alpha^F) \le \varepsilon_F^*\big\}$. Then for all $x, y \in \mathcal{K}_\alpha^F$ with $d_G(x, y) \le \varepsilon_F^*$:*

$$|h_F(x) - h_F(y)| \;\le\; M_F \cdot d_G(x, y). \tag{8}$$

Proof. Fix $x, y \in \mathcal{K}_\alpha^F$ with $d_G(x, y) \le \varepsilon_F^*$. Let $g^\star \in G$ achieve $d_G(x, y) = \|x - g^\star \cdot y\|$. Since $d_G(x, y) \le \varepsilon_F^*$, the segment from x to $g^\star \cdot y$ lies in the ε_F^*-neighbourhood of $\mathcal{K}_\alpha^F$. By the mean value theorem,

$$|h_F(x) - h_F(g^\star \cdot y)| \le M_F \cdot \|x - g^\star \cdot y\| = M_F \cdot d_G(x, y).$$

Using $h_F(g^\star \cdot y) = h_F(y)$ (G-invariance) completes the proof. $\square$

With the covering number (7), the ball-mass bound, and Lemma 3.8, the empirical oscillation theorem of Hu [2026b, Theorem 5.1] carries over to the quotient setting. The covering condition becomes

$$\Big(\frac{D_F}{\varepsilon} + 1\Big)^d \cdot \big(1 - (s/h_{\max})\, \pi_{\min}^D\, V_d\, \varepsilon^d\big)^{n_{\text{eff}}} \;\le\; \frac{\delta}{3}, \tag{9}$$

where $D_F = \mathrm{diam}(\mathcal{K}_\alpha^F)$, $\pi_{\min}^D = \inf_{\mathcal{K}_\alpha \cap D} \pi$, and $h_{\max} = \max_{z\in\mathcal{K}_\alpha^F} |\mathrm{Stab}(z)|$. Under Assumption 3.7, $h_{\max} = 1$ and the mass factor simplifies to $\pi_{\min}^F \cdot V_d \cdot \varepsilon^d$. The resulting quantile-core diagnostic is developed in Section 4.

Remark 3.9 (Computational scalability). The orbit sum in (4) has $s = |G|$ terms. For S_m with $m \le 5$ ($s \le 120$), this is inexpensive. For larger m, the sum can be approximated by restricting to permutations that lie within a neighbourhood of the identity, since distant permutations contribute negligibly when the modes are well separated.

4 Theoretical guarantees

We now establish that the oscillation framework of Hu [2026b] transfers to the quotient setting of Section 3, and that the resulting quantile-core convergence diagnostic is at least as sharp as—and typically much sharper than—the bound obtained on the original space.

4.1 The folded convergence diagnostic

The empirical oscillation theorem of Hu [2026b, Theorem 5.1] requires three regularity conditions on the target, the HPD set, and the log-density ratio. We verify each in turn for the quotient setting $(\pi_F, \mathcal{K}_\alpha^F, h_F)$.

Regularity condition 1 (Compact HPD set with positive density floor). By Proposition 3.2, $\mathcal{K}_\alpha^F = \mathcal{K}_\alpha \cap D$ is compact (intersection of a compact set and a closed domain). Its diameter satisfies $D_F \leq D_U := \operatorname{diam}(\mathcal{K}_\alpha)$ (Corollary 3.3a), and its density floor satisfies $\pi_{\min}^F \geq s\,\pi_{\min} > 0$ (Corollary 3.3b).

Regularity condition 2 (Smooth HPD boundary). The boundary $\partial\mathcal{K}_\alpha^F$ consists of three parts: (i) the HPD level set $\{U_F = u_\alpha^F\} \cap \operatorname{int}(D)$, which is a smooth $(d-1)$-dimensional manifold wherever $\nabla U_F \neq 0$ (holding generically by Sard's theorem); (ii) the fold boundary $\partial D \cap \operatorname{int}(\mathcal{K}_\alpha^F)$, which inherits the smoothness of ∂D (a hyperplane for reflections, a polyhedral cone for permutations); and (iii) the corner set $\{U_F = u_\alpha^F\} \cap \partial D$, which has Hausdorff dimension at most $d-2$ and does not affect the covering argument. Under Assumption 3.7, parts (ii) and (iii) are empty and $\partial\mathcal{K}_\alpha^F$ is smooth.

Regularity condition 3 (Local regularity of h_F). On $\operatorname{int}(D)$, $\pi_F = s\pi$ inherits the smoothness of π (typically C^∞ for Gaussian mixtures), and $q_F = \sum_g q_{T_F}(g\,\cdot\,)$ is C^∞ when T_F uses tanh activations. Therefore $h_F = \log \pi_F - \log q_F$ is C^1 on a neighbourhood of $\mathcal{K}_\alpha^F$, and the gradient supremum M_F (Definition in Lemma 3.8) is finite.

With all three conditions verified, we state the folded diagnostic.

Theorem 4.1 (FolT-MCMC convergence diagnostic). *Suppose π is G-invariant with $|G| = s$, Assumption 3.7 holds, and h_F is C^1 on a neighbourhood of $\mathcal{K}_\alpha^F$. Draw $Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} \pi_F$ and let $\widehat{\text{osc}}_n^F = \max_i h_F(Z_i) - \min_i h_F(Z_i)$. Let ε_F^* be the smallest $\varepsilon > 0$ satisfying (9) with $h_{\max} = 1$, and define $C_F = \widehat{\text{osc}}_n^F + 2M_F\varepsilon_F^*$.*

Then with probability at least $1 - \delta$:

$$\text{osc}_{\mathcal{K}_\alpha^F}(h_F) \;\leq\; C_F \;=\; \widehat{\text{osc}}_n^F + 2\, M_F\, \varepsilon_F^*. \tag{10}$$

The resulting α-HPD convergence diagnostic is

$$\underline{\gamma}_{F,\alpha} \;:=\; \frac{2}{1 + \exp(C_F)}. \tag{11}$$

We call $\underline{\gamma}_{F,\alpha}$ the HPD-core convergence diagnostic. In all experiments we report the quantile-core convergence diagnostic (denoted QC $\underline{\gamma}$) at $\rho = 0.05$, following Hu [2026a].

Remark 4.2 (Scope of the diagnostic). The bound (11) controls density-ratio stability on the HPD core $\mathcal{K}_\alpha^F$. It does not constitute a full-support spectral-gap certificate, which would require controlling $\sup_D h_F - \inf_D h_F$ including the tails where π_F has negligible mass. Full-support certification for transport MCMC with unbounded state spaces is an open problem shared with the frameworks of Hu [2026b,a]. The contribution of the present paper is to restore non-vacuousness of the core diagnostic in the presence of symmetric multimodality, not to close the core-to-full-support gap.

Proof. Apply Hu [2026b, Theorem 5.1] with the quotient-metric covering (Section 3.4): the covering number is bounded by (7), the ball mass by $\pi_{\min}^F \cdot V_d \cdot \varepsilon^d$ (Assumption 3.7), and the interpolation error by $M_F \cdot d_G$ (Lemma 3.8). Applying the Mengersen–Tweedie lower-bound functional [Mengersen and Tweedie, 1996] to the bounded HPD-core oscillation gives the core diagnostic (11). □

4.2 Covering radius improvement

The folded diagnostic (10) has the same *form* as the unfolded one, but its three constituent quantities are more favourable. We begin with the covering radius.

Theorem 4.3 (Covering radius improvement)**.** *Let ε_U^* denote the smallest $\varepsilon > 0$ satisfying the unfolded covering condition*

$$\left(\frac{D_U}{\varepsilon} + 1\right)^d \cdot \left(1 - \pi_{\min} V_d \, \varepsilon^d\right)^{n_{\text{eff}}} \le \frac{\delta}{3}, \tag{12}$$

and let ε_F^* be defined by (9) with $h_{\max} = 1$. Then $\varepsilon_F^* \le \varepsilon_U^*$.

Proof. Write the left-hand side of the covering condition as $\Phi(D_K, \pi_{\min}, \varepsilon) = (D_K/\varepsilon + 1)^d \, (1 - \pi_{\min} V_d \, \varepsilon^d)^{n_{\text{eff}}}$. For any fixed $\varepsilon > 0$:

1. $D_F \le D_U$ (Corollary 3.3a), so the first factor does not increase;
2. $\pi_{\min}^F \ge s \, \pi_{\min} \ge \pi_{\min}$ (Corollary 3.3b), so $\pi_{\min}^F V_d \varepsilon^d \ge \pi_{\min} V_d \varepsilon^d$, which makes the base of the second factor smaller and hence the second factor does not increase.

Therefore $\Phi(D_F, \pi_{\min}^F, \varepsilon) \le \Phi(D_U, \pi_{\min}, \varepsilon)$ for all $\varepsilon > 0$. Since ε_U^* satisfies $\Phi(D_U, \pi_{\min}, \varepsilon_U^*) \le \delta/3$, we have $\Phi(D_F, \pi_{\min}^F, \varepsilon_U^*) \le \delta/3$, so by minimality $\varepsilon_F^* \le \varepsilon_U^*$. The same monotonicity argument extends to the stabiliser-corrected case whenever $(s/h_{\max}) \, \pi_{\min}^D \ge \pi_{\min}$, which holds in the finite-group settings considered here. □

Remark 4.4 (Asymptotic scaling). Under simplifying approximations (dropping the $+1$ term and using $\log(1 - p) \approx -p$), the ratio $\varepsilon_F^*/\varepsilon_U^*$ scales heuristically as $s^{-1/d}$, up to logarithmic factors. This explains the empirical observation that the covering-radius improvement is more pronounced in low dimensions (Section 5). However, the rigorous content of Theorem 4.3 is the inequality $\varepsilon_F^* \le \varepsilon_U^*$, which is independent of the cross-mode misspecification condition.

4.3 Diagnostic improvement

The covering-radius improvement (Theorem 4.3) guarantees that the radius factor ε_F^* does not increase. The full product $M_F \varepsilon_F^*$ need not be smaller unless M_F is comparable to M_U, which is why the sufficient condition below keeps both terms explicit. The dominant source of improvement, however, is the empirical oscillation $\widehat{\operatorname{osc}}_n^F$, which drops when the quotient proposal eliminates cross-mode residuals.

We now compare the folded and unfolded diagnostics. Let $T_U : \mathbb{R}^d \to \mathbb{R}^d$ be a normalising flow trained on the original (unfolded) target π, with proposal density $q_U = (T_U)_\# \mathcal{N}(0, I)$ and log-density ratio $h_U = \log \pi - \log q_U$. The unfolded diagnostic is $C_U = \widehat{\operatorname{osc}}_n + 2M_U \varepsilon_U^*$, where $\widehat{\operatorname{osc}}_n$, M_U, ε_U^* are defined analogously to their folded counterparts but using h_U and $\mathcal{K}_\alpha$.

To quantify the oscillation gap we introduce three conditions.

(M′) Relative cross-mode deficiency. For each off-mode component $\mathcal{K}_\alpha^{(j)}$, $j \neq 1$, there exists $\Delta_j > 0$ such that

$$\inf_{\theta \in \mathcal{K}_\alpha^{(j)}} h_U(\theta) \;\geq\; \sup_{\theta \in \mathcal{K}_\alpha^{(1)}} h_U(\theta) \;+\; \Delta_j. \tag{13}$$

(W) Folded within-mode fit. There exists $r_1 \geq 0$ such that

$$\operatorname{osc}_{\mathcal{K}_\alpha^F}\big(\log(s\pi) - \log q_{T_F}\big) \;\leq\; 2r_1. \tag{14}$$

(R) Folded-proposal residual.

$$r_F \;:=\; \sup_{z \in \mathcal{K}_\alpha^F} \big|\log q_F(z) - \log q_{T_F}(z)\big| \;<\; \infty. \tag{15}$$

Condition (M′) says the unfolded log-density ratio h_U is uniformly higher on every off-mode than on the primary mode: the unfolded flow under-covers those regions by at least a factor of e^{Δ_j}. This holds whenever the spectrally normalised RealNVP has insufficient capacity to represent multiple separated modes—a common situation in practice, as illustrated by the experiments in Section 5 and explicitly checked in representative cases in Supplement D.5. Condition (W) bounds the within-mode oscillation of the folded flow's unnormalised log-density ratio on $\mathcal{K}_\alpha^F$. Condition (R) quantifies the contribution of off-mode orbit terms $q_{T_F}(g \cdot z)$ for $g \neq e$; for well-separated modes $r_F \approx 0$ since these terms are exponentially small.

Note that (M′) involves the unfolded flow T_U while (W) and (R) involve the folded flow T_F; this reflects the two-flow comparison at the heart of FolT-MCMC.

Lemma 4.5 (Oscillation gap)**.** *Under conditions* (M′), (W)*, and* (R)*, the empirical oscillations on the unfolded and folded spaces satisfy, with probability at least* $1-\delta'$*,*

$$\widehat{\mathrm{osc}}_n \;\ge\; \widehat{\mathrm{osc}}_n^F \;+\; \min_{j\neq 1}\Delta_j \;-\; 2r_1 - r_F \;-\; c_n, \tag{16}$$

where $c_n = O\big(\sqrt{\log(2/\delta')/n}\big)$ *accounts for the sampling error in both empirical oscillations.*

Proof. We bound $\widehat{\mathrm{osc}}_n$ from below and $\widehat{\mathrm{osc}}_n^F$ from above.

Lower bound on $\widehat{\mathrm{osc}}_n$*.* Condition (M′) gives, for any off-mode $j \neq 1$, $\inf_{\mathcal{K}_\alpha^{(j)}} h_U \ge \sup_{\mathcal{K}_\alpha^{(1)}} h_U + \Delta_j$. Therefore

$$\mathrm{osc}_{\mathcal{K}_\alpha}(h_U) \;\ge\; \inf_{\mathcal{K}_\alpha^{(j)}} h_U - \inf_{\mathcal{K}_\alpha^{(1)}} h_U \;\ge\; \sup_{\mathcal{K}_\alpha^{(1)}} h_U + \Delta_j - \inf_{\mathcal{K}_\alpha^{(1)}} h_U \;\ge\; \min_{j\neq 1}\Delta_j.$$

The empirical oscillation $\widehat{\mathrm{osc}}_n$ concentrates around $\mathrm{osc}_{\mathcal{K}_\alpha}(h_U)$ up to a term $O(\sqrt{\log/n})$.

Upper bound on $\widehat{\mathrm{osc}}_n^F$*.* For $z \in \mathcal{K}_\alpha^F = \mathcal{K}_\alpha^{(1)} \cap D$ (a.e.),

$$h_F(z) = \log(s\pi(z)) - \log q_F(z) = \underbrace{\log(s\pi(z)) - \log q_{T_F}(z)}_{\text{within-mode: osc}\le 2r_1} \;-\; \log\Big(1 + \frac{\sum_{g\neq e} q_{T_F}(gz)}{q_{T_F}(z)}\Big).$$

The last term lies in $[-r_F, 0]$ by condition (R). Therefore $\mathrm{osc}_{\mathcal{K}_\alpha^F}(h_F) \le 2r_1 + r_F$, and $\widehat{\mathrm{osc}}_n^F$ concentrates similarly.

Combining: $\widehat{\mathrm{osc}}_n - \widehat{\mathrm{osc}}_n^F \ge \min_j \Delta_j - 2r_1 - r_F - c_n$. □

We can now state the main comparison result.

Theorem 4.6 (Diagnostic improvement)**.** *Adopt the setting of Theorem 4.1 and let* $C_U = \widehat{\mathrm{osc}}_n + 2M_U\varepsilon_U^*$ *be the unfolded diagnostic (with gradient constant* M_U *and covering* *radius ε_U^*). Suppose conditions* (M′), (W), (R) *hold, and define*

$$\Delta \;=\; \min_{j\neq 1} \Delta_j - 2r_1 - r_F - c_n.$$

If

$$\Delta \;>\; 2\big(M_F\,\varepsilon_F^* \;-\; M_U\,\varepsilon_U^*\big)^+ \tag{17}$$

(where $(x)^+ = \max(x,0)$), then with probability at least $1-\delta_U-\delta_F-\delta'$:

$$C_F \;<\; C_U, \qquad \frac{\underline{\gamma}_{F,\alpha}}{\underline{\gamma}_{U,\alpha}} \;=\; \frac{1+\exp(C_U)}{1+\exp(C_F)} \;>\; 1, \tag{18}$$

where $\underline{\gamma}_{F,\alpha}$ and $\underline{\gamma}_{U,\alpha}$ are the respective HPD-core convergence diagnostics.

Proof. Write

$$\begin{aligned} C_U - C_F &= \big(\widehat{\mathrm{osc}}_n - \widehat{\mathrm{osc}}_n^F\big) + 2\big(M_U\varepsilon_U^* - M_F\varepsilon_F^*\big) \\ &\geq \Delta + 2\big(M_U\varepsilon_U^* - M_F\varepsilon_F^*\big) \qquad \text{(Lemma 4.5)}. \end{aligned}$$

If $M_F\varepsilon_F^* \leq M_U\varepsilon_U^*$, both terms are non-negative and $C_U - C_F \geq \Delta > 0$. If $M_F\varepsilon_F^* > M_U\varepsilon_U^*$, condition (17) gives $\Delta > 2(M_F\varepsilon_F^* - M_U\varepsilon_U^*)$, so again $C_U - C_F > 0$. The convergence-diagnostic comparison follows from the monotonicity of $C \mapsto 2/(1+e^C)$. The confidence level combines the three events via a union bound: the folded diagnostic (prob $\geq 1-\delta_F$), the unfolded diagnostic ($\geq 1-\delta_U$), and the oscillation-gap concentration ($\geq 1-\delta'$). □

4.4 When does folding help?

Theorems 4.3 and 4.6 identify two complementary improvement mechanisms: a covering-radius reduction that holds independently of the cross-mode deficiency condition ($\varepsilon_F^* \leq \varepsilon_U^*$, under Assumption 3.7) and an oscillation reduction that holds under condition (M′).

The role of (M′). Condition (M′) is *not* an assumption about the target; it is a condition on the *unfolded flow* T_U. It asserts that the trained unfolded transport map fails to cover at least one off-mode. This holds naturally for spectrally normalised RealNVP with scale clipping, whose limited expressiveness makes faithful representation of multiple separated modes difficult—especially in higher dimensions, where the coupling architecture must mediate inter-coordinate information through the conditioner network. In representative experiments (Supplement D.5), condition (M′) is strongly satisfied, with Δ/C_U between 0.87 and 0.90.

When the flow *is* expressive enough to represent all modes faithfully, $\Delta \approx 0$ and the oscillation gap vanishes; the improvement then comes solely from the covering-radius reduction (Theorem 4.3), which is modest—heuristically $O(s^{-1/d})$ (Remark 4.4).

Fold boundary placement. The main-text results (Theorems 4.1–4.6) are stated under Assumption 3.7, i.e. the HPD set does not touch the fold boundary. The boundary-touching case is handled by the stabiliser-corrected covering condition (9) and is detailed in Supplement C. When the fold boundary passes through a high-density region—as in our banana diagnostic experiment (Section 5.2)—the full oscillation can increase due to boundary residuals, even while the quantile-core diagnostic improves. This provides a practical design principle: *the fold boundary should lie in a low-density region separating the symmetric modes.* For label-switching mixtures, the sorted-chamber boundary is automatically in the valley between modes whenever the components are well separated.

Dimension dependence. A key empirical finding is that QC $\underline{\gamma}_F$ is empirically nearly dimension-free: it stays in $[0.90, 0.94]$ across $d = 2$ to 20 for Gaussian mixtures, while QC $\underline{\gamma}_U$ collapses from 0.40 to 0.016 (Section 5.3). The theoretical explanation is that all three terms in C_F remain small as d grows: $\widehat{\mathrm{osc}}_n^F$ reflects only the single-mode transport residual, M_F is controlled by spectral normalisation, and ε_F^* benefits from the s-fold density floor. In contrast, C_U is dominated by the cross-mode oscillation, which grows with d because the coupling-layer architecture struggles increasingly to generate multimodal structure in high dimensions.

Mode count vs. dimension. The label-switching experiments (Section 5.4) reveal that the unfolded diagnostic degrades primarily with the number of modes $s = m!$, not with the dimension d. Comparing $m = 3, p = 2$ ($d = 6$, $s = 6$) with $m = 3, p = 4$ ($d = 12$, $s = 6$): doubling the dimension while keeping s fixed reduces QC $\underline{\gamma}_U$ by only a factor of 2, whereas increasing s from 6 to 24 (at fixed block size) reduces it by a factor of 15. The folded diagnostic varies much less across both changes.

5 Experiments

We evaluate FolT-MCMC on three synthetic target families designed to isolate the mechanisms predicted by the theory, a comparison with the random permutation sampler [Frühwirth-Schnatter, 2001], and a standard Bayesian Gaussian mixture posterior. The first experiment is a diagnostic stress test showing that folding can be harmful when the fold boundary intersects high-density mass. The second evaluates dimension scaling for a well-separated reflection-symmetric Gaussian mixture. The third evaluates permutation folding for label-switching mixtures. The fourth compares FolT-MCMC with the standard label-switching baseline. The fifth applies all methods to a textbook Bayesian three-component mixture posterior. Across experiments, we report the quantile-core convergence diagnostic, NLL, acceptance, and effective sample size (ESS). All experiments use the same architecture and training protocol; the only variable across conditions is whether folding is applied.

5.1 Setup

Architecture. Both the folded and unfolded flows are spectrally normalised RealNVP [Dinh et al., 2017, Miyato et al., 2018] with tanh activations and scale clipping at $c = 0.7$, following Hu [2026b]. Layer count and hidden dimension scale with d: $(L, h) = (8, 64)$ for $d \leq 4$; $(10, 128)$ for $d = 5$–6; $(12, 128)$ for $d = 8$–12; $(16, 256)$ for $d = 20$. The folded and unfolded pipelines share identical architecture for each d.

Training. Negative log-likelihood with annealed oscillation and gradient regularisation (FolT-OG-Anneal), as in Hu [2026a]. Training uses Adam with learning rate 10^{-3}, batch size 512, and 2 000 epochs for $d \leq 6$ (increasing to 3 500 for $d \geq 8$). Training samples are drawn from the exact sampler of each target (folded samples projected onto D).

Diagnostic computation. The quantile-core convergence diagnostic (denoted QC $\underline{\gamma}$) at $\rho = 0.05$ (primary) and $\rho = 0.025$ (secondary), with $n = 20\,000$ independent diagnostic samples and Dvoretzky–Kiefer–Wolfowitz (DKW)-corrected confidence $\delta = 0.05$, following the protocol of Hu [2026a].

MCMC. Independence Metropolis–Hastings with 4 chains of length 5 000 (burn-in 1 000). Effective sample size estimated via batch means.

Hardware and code. NVIDIA RTX 4090 GPU; code will be released upon acceptance.

5.2 Diagnostic: fold boundary location

Purpose. To demonstrate that the fold boundary's position relative to the target's density mass is a critical design variable, and that folding can *hurt* the full oscillation bound when the boundary passes through a high-density region.

Target. Asymmetric double banana in $d = 2$: a mixture of two banana-shaped modes with curvatures $a = 1.0$ and $b = 0.5$, related by reflection $\theta_1 \mapsto -\theta_1$. Because $a \neq b$, the target is not strictly $\mathbb{Z}_2$-invariant; this experiment intentionally stresses the method outside the exact group-invariant setting of Section 3. The fold boundary $\{\theta_1 = 0\}$ passes through the high-density region near the origin where both modes overlap.

Results. Table 1 summarises the comparison.

The full oscillation *increases* under folding (58.4 vs. 30.7), because the smooth flow cannot produce the hard truncation at $\theta_1 = 0$ where the target density is high, creating a boundary residual spike. Nevertheless, the quantile-core convergence diagnostic still improves: QC $\underline{\gamma}$ rises from 0.272 to 0.320 at $\rho = 0.05$ and from 0.064 to 0.135 at $\rho = 0.025$, because the ρ-trimming excludes the boundary spike. The NLL drops by 23% ($4.34 \to 3.34$), confirming that the flow fits the single folded mode more easily.

Table 1: Banana diagnostic ($d = 2$, $s = 2$). The fold boundary passes through a high-density region.

Metric	Unfolded	Folded
Full oscillation	30.7	58.4
QC oscillation ($\rho = 0.05$)	1.38	1.10
QC $\underline{\gamma}$ ($\rho = 0.05$)	0.272	0.320
QC $\underline{\gamma}$ ($\rho = 0.025$)	0.064	0.135
Final NLL	4.34	3.34
ESS / sample	0.048	0.042
Acceptance rate	0.604	0.594

Table 2: Dimension scaling for Gaussian mixture ($s = 2$, $\rho = 0.05$).

d	QC $\underline{\gamma}_U$	QC $\underline{\gamma}_F$	Ratio	QC osc_U	QC osc_F
2	0.402	0.936	2.3×	1.38	0.13
5	0.094	0.941	10.0×	3.01	0.12
10	0.016	0.922	59.2×	4.85	0.16
20	0.016	0.902	57.0×	4.83	0.20

This experiment validates the design principle of Section 4.4: *folding helps when the fold boundary lies in a low-density valley, not when it cuts through the target's mass.* The subsequent experiments use targets satisfying this condition.

5.3 Dimension scaling: well-separated mixture

Purpose. To test whether the folded diagnostic is robust to increasing dimension, in contrast to the unfolded diagnostic.

Target. Symmetric two-component Gaussian mixture, $\pi = \frac{1}{2}\mathcal{N}(\mu_1, I_d) + \frac{1}{2}\mathcal{N}(\mu_2, I_d)$, with $\mu_1 = (3, 0, \ldots, 0)$, $\mu_2 = -\mu_1$ (separation $\delta = 6$). Reflection fold with $G = \mathbb{Z}_2$, $s = 2$. Dimensions $d \in \{2, 5, 10, 20\}$. The fold boundary $\{\theta_1 = 0\}$ sits in the inter-mode valley where the density relative to a single-component peak is $\exp(-9/2) \approx 0.011$.

Results. Table 2 and Figure 2 present the key comparison.

The folded quantile-core diagnostic QC $\underline{\gamma}_F$ remains in $[0.90, 0.94]$ across $d = 2$ to 20, while the unfolded diagnostic collapses from 0.40 to 0.016—a 59-fold gap by $d = 10$. The quantile-core oscillation bound tells the same story: folded QC oscillation stays below 0.20 at all dimensions, while the unfolded QC oscillation grows from 1.38 to 4.85.

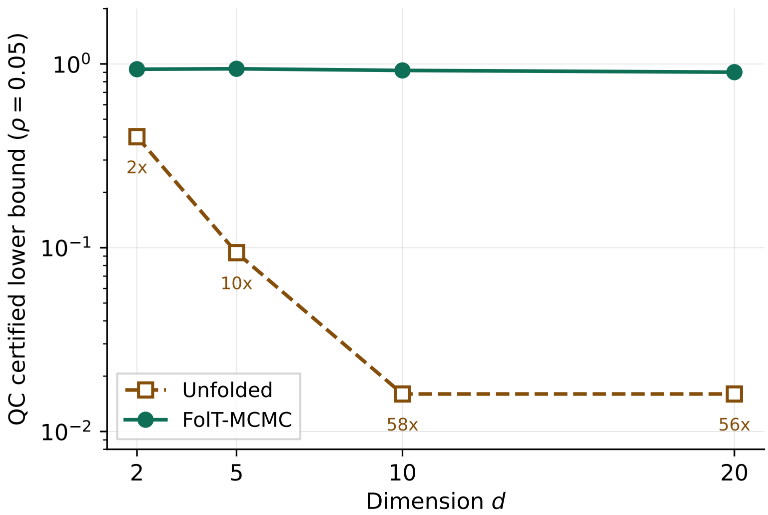

Figure 2: Quantile-core convergence diagnostic vs. dimension (Gaussian mixture, $s = 2$, $\rho = 0.05$). The folded diagnostic is empirically nearly dimension-free; the unfolded diagnostic collapses. Numbers indicate the improvement ratio at each dimension.

At $d = 20$, the folded NLL of 28.48 is within 0.1 of the entropy of a d-dimensional standard Gaussian, $\frac{d}{2}\log(2\pi e) \approx 28.4$, indicating that the folded transport achieves a near-optimal fit. The unfolded flow, by contrast, must represent a bimodal target in 20 dimensions using a coupling architecture that can only influence each coordinate conditionally—a task at which it demonstrably fails.

5.4 Label-switching scaling

Purpose. To verify that permutation folding gives the same strong improvement as reflection folding, and to disentangle the effects of mode count s and dimension d on the unfolded diagnostic.

Target. Label-switching Gaussian mixtures with m components, each a block of p parameters (representing, e.g., frequency and damping of a structural mode), total dimension $d = mp$. Component centres are spaced by 3 standard deviations along the sort coordinate. Permutation fold with $G = S_m$, $s = m!$. Four configurations are tested (Table 3).

Results. The folded quantile-core diagnostic QC $\underline{\gamma}_F$ stays in $[0.62, 0.82]$ across all configurations, while the unfolded diagnostic collapses from 0.366 to 0.0043 as the number of

Table 3: Label-switching scaling (quantile-core, $\rho = 0.05$).

Config	d	s	QC $\underline{\gamma}_U$	QC $\underline{\gamma}_F$	Ratio	QC osc_U	QC osc_F
m=2, p=2	4	2	0.366	0.820	2.2×	1.50	0.37
m=3, p=2	6	6	0.066	0.685	10.4×	3.38	0.65
m=3, p=4	12	6	0.036	0.683	19.1×	4.01	0.66
m=4, p=2	8	24	0.0043	0.624	145×	6.14	0.79

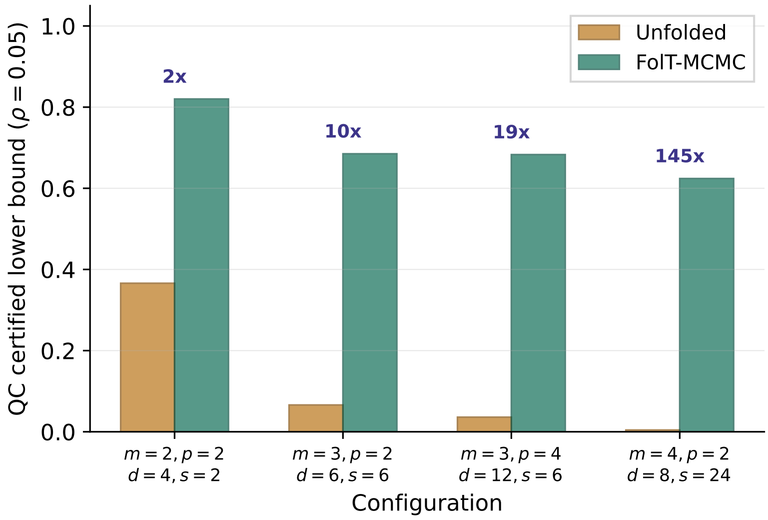

Figure 3: Quantile-core convergence diagnostic across label-switching configurations. The folded diagnostic varies slowly; the unfolded diagnostic collapses superlinearly with s. Numbers indicate the improvement ratio.

equivalent modes grows from $s = 2$ to $s = 24$. The improvement ratio reaches 145× at $m = 4$.

Disentangling mode count from dimension. Comparing $m = 3, p = 2$ ($d = 6$, $s = 6$, QC $\underline{\gamma}_U = 0.066$) with $m = 3, p = 4$ ($d = 12$, $s = 6$, QC $\underline{\gamma}_U = 0.036$): doubling the dimension at fixed s reduces QC $\underline{\gamma}_U$ by a factor of ≈ 2. Comparing $m = 3, p = 2$ ($s = 6$) with $m = 4, p = 2$ ($s = 24$): increasing s by 4× at comparable d reduces QC $\underline{\gamma}_U$ by a factor of ≈ 15. In this controlled comparison, the unfolded diagnostic appears more sensitive to the number of symmetric copies than to the ambient dimension—consistent with the theoretical prediction of Section 4.4. The folded diagnostic varies much less across both changes (QC $\underline{\gamma}_F \in [0.68, 0.68]$ for the two $s = 6$ configurations despite d doubling).

Table 4: Baseline comparison ($m = 3$, $d = 6$, $s = 6$). Sorted ESS is computed after projecting all samples into the sorted chamber.

Method	Sorted ESS / sample	Acceptance	QC $\underline{\gamma}$
Unfolded IMH	0.607	0.622	0.066
FolT-MCMC	0.540	0.881	0.685
RPS	0.482	0.612	N/A

Convergence at $m = 4$. With $s = 24$ equivalent modes, the unfolded flow trains stably (no divergence or numerical issues) but fails representationally: NLL = 15.0, acceptance rate = 0.41, full-oscillation diagnostic $\gamma_{\text{full}} \approx 5 \times 10^{-10}$ (completely vacuous). Scatter plots confirm that the unfolded proposal spreads mass across the 24 symmetric chambers but yields a highly uneven density ratio, leading to low acceptance and a vacuous diagnostic. The folded chain, by contrast, samples efficiently in the sorted chamber (acceptance = 0.90, ESS/sample = 0.85).

5.5 Comparison with random permutation sampler

Purpose. To compare FolT-MCMC with the random permutation sampler (RPS) of Frühwirth-Schnatter [2001], the most widely used label-switching mitigation in Bayesian mixture modelling.

Setup. All three methods use the $m = 3$, $p = 2$ label-switching target ($d = 6$, $s = 6$) from Section 5.4. The unfolded IMH and RPS share the same trained flow as proposal; RPS additionally applies a random component-label permutation after each MH accept/reject step. FolT-MCMC uses its own folded flow in the sorted chamber. Chains: $4 \times 10\,000$ steps, burn-in $2\,000$. To ensure a fair mixing comparison, we report ESS after projecting all samples into the sorted fundamental domain.

Results. The sorted ESS per sample is comparable across all three methods (0.48–0.61), well within the variance of the batch-means estimator. This is expected: all three use a global independence proposal from a well-trained flow that already covers all six modes, so no chain gets stuck in a single mode—the setting where RPS was designed to help [Frühwirth-Schnatter, 2001]. (The per-sample permutation step in our RPS implementation is not batched, but wall-clock time is not the basis of this comparison.)

The key distinction is structural rather than computational. FolT-MCMC's folded diagnostic QC $\underline{\gamma} = 0.685$ predicts ESS$/n \approx \gamma/(2-\gamma) = 0.52$, matching the empirical 0.54—a tight diagnostic. The unfolded diagnostic QC $\underline{\gamma} = 0.066$ predicts ESS$/n \approx 0.034$, far below the empirical 0.61: the diagnostic is valid but loose, reflecting worst-case cross-mode oscillation that did not materialise on this particular run. The random permutation sampler can improve label exploration, but its composite IMH-plus-permutation kernel does not fit the independence-proposal minorisation structure that the convergence diagnostic requires. FolT-MCMC, by contrast, admits a non-vacuous diagnostic because it samples from a single-mode target with a learned proposal.

This comparison underscores the core thesis of FolT-MCMC: *folding does not primarily improve raw mixing speed; it restores a non-vacuous convergence diagnostic for the chain*, closing the gap between the provable lower bound and the empirical performance.

5.6 Standard Bayesian mixture posterior

Purpose. To validate FolT-MCMC on a textbook Bayesian inference problem rather than a method-specific synthetic target, and to demonstrate that a non-vacuous convergence diagnostic requires quotient-space sampling even when post-hoc relabelling recovers correct point estimates. This separates point-estimation recovery from a reliable convergence diagnostic.

Model. $N = 500$ observations drawn from a three-component univariate Gaussian mixture with equal weights $w = (1/3, 1/3, 1/3)$ and unknown component means and standard deviations. Parameters: $\theta = (\mu_1, \log\sigma_1, \mu_2, \log\sigma_2, \mu_3, \log\sigma_3)$, $d = 6$. Priors: $\mu_k \sim \mathcal{N}(3, 5^2)$, $\log\sigma_k \sim \mathcal{N}(0, 0.5^2)$. The equal weights and exchangeable priors give the posterior exact S_3 label-switching symmetry ($s = 6$). Training samples are generated by a random-walk Metropolis pre-run of 200 000 steps, thinned to 50 000.

Methods compared. Four pipelines, all using the same RealNVP architecture (10 layers, hidden dim 128): (i) unfolded IMH; (ii) FolT-MCMC with permutation fold ($D =$

$\{\mu_1 \leq \mu_2 \leq \mu_3\}$); (iii) RPS (unfolded flow + random permutation); (iv) post-hoc sort (unfolded chain samples sorted by μ_k).

Results. The unfolded RealNVP cannot fit the six equally weighted, well-separated posterior modes (full oscillation = 443), so the unfolded IMH chain freezes entirely: acceptance rate $\approx$0, and the raw labelled parameter estimates ($\hat{\mu} = 1.15, 5.60, 2.07$, not sorted) reflect the single mode in which the chain is stuck rather than the true posterior. The diagnostic is completely vacuous (QC $\underline{\gamma} \approx 10^{-27}$).

FolT-MCMC, sampling in the sorted chamber, achieves acceptance 0.960 and recovers the true parameters accurately: $\hat{\mu} = (-0.12, 2.95, 5.98)$, $\hat{\sigma} = (0.78, 0.92, 0.81)$. The quantile-core diagnostic QC $\underline{\gamma} = 0.878$ is the only non-vacuous diagnostic among the four methods.

Post-hoc relabelling: correct point estimates, no guarantee. Under exact S_3 symmetry with equal weights, all six posterior modes have identical shape. Consequently, post-hoc sorting of a frozen chain yields correct sorted point estimates ($\hat{\mu} = -0.16, 2.99, 5.99$), because sorting any single mode's samples recovers the true ordered parameters regardless of which mode the chain occupies. However, the chain has not explored the posterior: it provides no reliable uncertainty quantification and only a vacuous convergence diagnostic (QC $\underline{\gamma} \approx 0$). This reveals a subtle but important distinction: *post-hoc relabelling can recover correct point estimates under exact symmetry, but it cannot verify that the sampler has converged or that credible intervals are valid.* Quotient-space sampling is needed not for point estimation per se, but for diagnosing posterior convergence.

RPS: correct estimates, no diagnostic. The random permutation sampler, using the same frozen flow but forcing label permutations, achieves textbook label switching (visible in the raw trace, Figure 4 centre) and recovers accurate sorted estimates ($\hat{\mu} = -0.11, 3.02, 5.99$). However, its MH acceptance rate remains 0.001 (the permutation step does not improve proposal quality), and it provides no non-vacuous convergence

Table 5: Bayesian mixture benchmark (K=3, d=6, s=6, N=500, $w = (1/3, 1/3, 1/3)$). True parameters: $\mu = (0, 3, 6)$, $\sigma = (0.8, 1.0, 0.7)$. Estimates $\hat{\mu}_k$ are sorted posterior means for all methods except unfolded IMH, which reports raw labelled means from the single mode in which the chain is stuck.

Method	QC $\underline{\gamma}$	Accept	$\hat{\mu}_1$	$\hat{\mu}_2$	$\hat{\mu}_3$
Unfolded IMH	≈ 0	≈ 0	1.15	5.60	2.07
FolT-MCMC	0.878	0.960	-0.12	2.95	5.98
RPS	N/A	0.001	-0.11	3.02	5.99
Post-hoc sort	$\approx 0^{\dagger}$	≈ 0	-0.16	2.99	5.99

† Inherited from the unfolded chain; post-hoc sorting does not define an independent sampler.

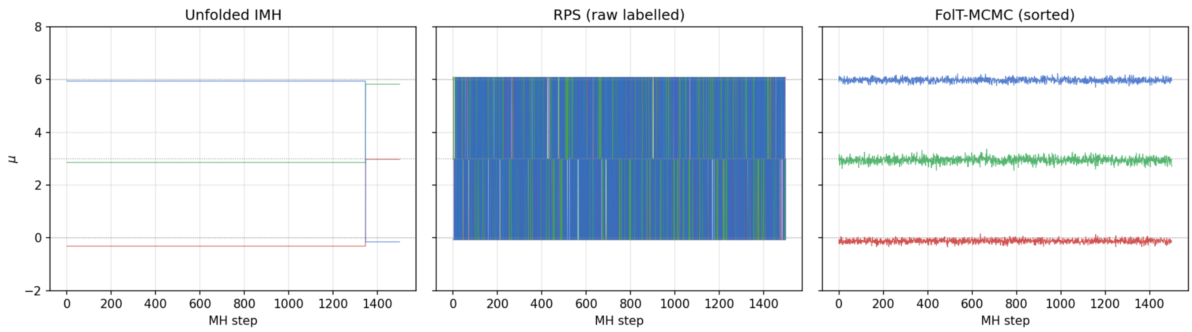

Figure 4: Trace plots for the Bayesian mixture posterior. Left: unfolded IMH is frozen (acceptance ≈ 0). Centre: RPS exhibits textbook label switching in raw labelled coordinates. Right: FolT-MCMC in sorted space shows clean separated bands.

diagnostic.

6 Application: closely-spaced structural modes

We apply FolT-MCMC to Bayesian modal identification of a supertall building instrumented during Typhoon Mangkhut (2018). Three closely-spaced modes in the 0.7–1.1 Hz band create an S_3 label-switching problem ($s = 6$) that renders the unfolded diagnostic vacuous.

6.1 Data and model

Data. Accelerometer recordings from 15 usable channels at 32 Hz. We select a single 30-minute window near peak typhoon intensity (band signal-to-noise ratio ≈ 18 dB) and compute the trace of the 15×15 cross-power spectral density matrix over 0.7–1.1 Hz (51 frequency ordinates), capturing the total vibrational power across all channels. Build-

Table 6: Structural modal identification ($d = 6$, $s = 6$, Typhoon Mangkhut window).

Method	QC $\underline{\gamma}$ (ρ=0.05)	Acceptance	ESS / sample	Label-switch rate
Unfolded	9.9×10^{-6} (vacuous)	0.204	0.059	0.170
FolT-MCMC	0.0018	0.371	0.080	0.000

ing identity and location are anonymised per the owner's request.

Model. A simplified Whittle likelihood with three single-degree-of-freedom Lorentzian peaks plus a flat noise floor. Each mode j has two parameters (natural frequency f_j, damping ratio ξ_j), giving $d = 6$ total. The overall amplitude is profiled out in closed form (conditional MLE), preserving exact S_3 symmetry. Uniform priors: $f_j \in [0.7, 1.1]\,\text{Hz}$, $\xi_j \in [0.001, 0.05]$.

This is deliberately simpler than a full multi-output Bayesian operational modal analysis: the goal is to exhibit the label-switching phenomenon and demonstrate FolT-MCMC's convergence-diagnostic advantage, not to compete on estimation accuracy.

Folding. Permutation fold with $G = S_3$, $s = 6$. The fundamental domain is the sorted chamber $D = \{f_1 \leq f_2 \leq f_3\}$. Parameters are block-whitened (one shared mean and standard deviation for the frequency slots, one for the damping slots) to bring all coordinates to unit scale while preserving the commutativity of whitening and sorting.

6.2 Results

Training uses 50 000 pre-samples from a long-run random-walk Metropolis chain, folded onto D. Architecture and diagnostic protocol follow Section 5.1.

Label switching. The unfolded chain exhibits dense label switching, with 17% of steps swapping the frequency assignment among the three modes (Table 6). The FolT-MCMC chain in sorted space shows zero label switches: the sorted frequencies remain cleanly separated throughout the run.

Convergence diagnostic. The unfolded quantile-core diagnostic is QC $\underline{\gamma} \approx 10^{-5}$—completely vacuous, as the six permutation modes inflate the oscillation bound beyond

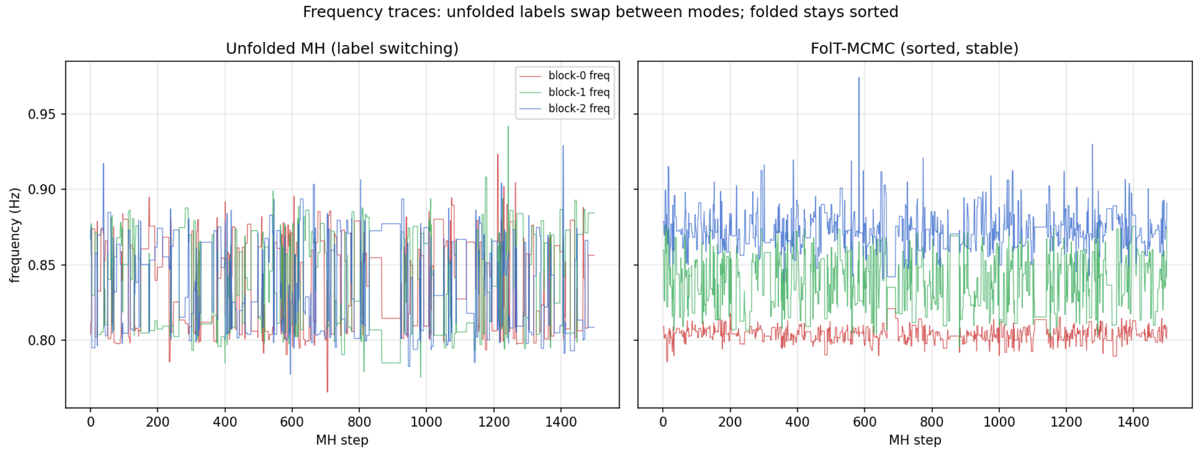

Figure 5: Trace plots of the three identified frequencies. Left: unfolded chain with frequent label switching (17% of steps). Right: FolT-MCMC in sorted space with no label switching.

any useful range. FolT-MCMC provides a non-vacuous quantile-core diagnostic (QC $\underline{\gamma} = 0.0018$), a 180× improvement. The absolute value is modest, reflecting the genuine difficulty of closely-spaced modes: the near-degenerate boundary $f_i \approx f_j$ in the sorted chamber creates heavy-tailed density ratios that resist a tight diagnostic. Nevertheless, the qualitative transition from *no usable diagnostic* to *a usable diagnostic* is the operationally meaningful distinction.

Posterior estimates. The posterior median frequencies are $\hat{f}_1 = 0.804$, $\hat{f}_2 = 0.838$, $\hat{f}_3 = 0.873$ Hz, recovering the closely-spaced cluster structure. These values are approximately 0.05 Hz below the nominal multi-output estimates from a separate full Bayesian OMA analysis of the same dataset [Hu and Lam, 2026], a systematic offset attributable to the single-output trace-spectrum model, which under-weights the transverse mode (≈ 0.93 Hz) relative to the translational modes. The demonstration targets the sampler's convergence diagnostic, not estimation accuracy; the full AD-BOMA model is developed separately [Hu and Lam, 2026].

7 Discussion

7.1 Design principles

The experiments point to a simple but consequential design rule: *the fold boundary should lie in a low-density valley between the symmetric modes.* When this condition holds (Gaussian mixture, label-switching targets), folding yields large diagnostic gains across all metrics. When it is violated (banana diagnostic with fold boundary through the density ridge), the full oscillation bound can increase, though the quantile-core diagnostic still improves modestly.

For well-separated label-switching mixtures, the sorted-chamber boundary $\{\theta_1^{(j)} = \theta_1^{(j+1)}\}$ lies in a low-density valley between adjacent mode centres. For closely-spaced structural modes, however, the boundary may carry non-negligible posterior mass; this explains why the structural example in Section 6 yields a non-vacuous but modest diagnostic (QC $\underline{\gamma} = 0.0018$).

7.2 Relation to existing work

Equivariant normalising flows. Köhler et al. [2020] and Rezende et al. [2019] construct flows satisfying $T(g \cdot z) = g \cdot T(z)$, encoding the symmetry into the architecture. This reduces the number of learnable parameters and ensures that the flow respects the group structure, but does not reduce the number of modes: the equivariant flow must still represent all $|G|$ copies. FolT-MCMC takes the dual approach of collapsing the $|G|$ copies into one via the quotient map. The two strategies are complementary: one could train an equivariant flow and then fold, combining parameter efficiency with mode reduction.

Post-hoc relabelling. Methods such as pivotal reordering [Stephens, 2000], equivalence-class representatives [Papastamoulis and Iliopoulos, 2010], and random permutation samplers [Frühwirth-Schnatter, 2001] operate after MCMC has been run. They improve the interpretability of the samples but do not address the mixing difficulty of the underlying chain. In particular, standard post-hoc relabelling methods do not provide the oscillation-based convergence diagnostic considered here. FolT-MCMC resolves the mixing problem at source by eliminating the symmetric modes before sampling, and inherits its convergence diagnostic from the framework of Hu [2026b].

Tempering and replica exchange. Parallel tempering [Geyer, 1991, Earl and Deem, 2005] overcomes multimodality by introducing heated replicas that can cross energy barriers. This is a general-purpose strategy that does not exploit symmetry. FolT-MCMC exploits the *structure* of the multimodality (it arises from a known group action) to remove it geometrically, without the overhead of maintaining multiple temperature levels.

7.3 Limitations and future work

Exact symmetry. The current framework requires the target to be exactly G-invariant. The simplified exchangeable model in Section 6 preserves exact label symmetry by construction (identical Lorentzian parameterisation and profiled amplitude). In more realistic Bayesian operational modal analysis models, however, mode-specific priors, spatial mode-shape information, or physical ordering constraints may break this symmetry approximately. Extending FolT-MCMC to ε-approximate group invariance, where $\|\pi(g\theta) - \pi(\theta)\|$ is small but nonzero, is a natural direction. We expect the quotient diagnostic to degrade gracefully with the symmetry-breaking magnitude, but a formal perturbation bound remains to be established.

Unknown symmetry groups. FolT-MCMC presupposes knowledge of G and of a fundamental domain D. For the applications considered here (mixture models, structural identification), the symmetry is known by construction. When the symmetry is latent or only partially known, a *learned folding map* could be trained jointly with the transport flow, analogous to recent work on learning group structure from data [Dehmamy et al., 2021].

Continuous symmetries. Finite groups cover label switching and discrete reflections, but some inference problems have continuous symmetries (rotation invariance, gauge symmetry in physics). Extending the quotient construction to compact Lie groups would require replacing the finite orbit sum in the quotient proposal with a Haar-measure integral, and the covering theory with manifold covering bounds.

Scalability. The orbit sum in the quotient proposal has $|G|$ terms. For S_m with moderate m ($m \leq 5$, $|G| \leq 120$), this is inexpensive. For larger groups, truncation or importance sampling over the orbit may be needed.

Core diagnostic vs. full-support guarantee. The quantile-core diagnostic reported throughout this paper is a density-ratio bound on the high-probability posterior core, not a full-support spectral-gap guarantee. This limitation is shared with the frameworks of Hu [2026b] and Hu [2026a]. A full-support certificate for transport MCMC with unbounded state spaces remains an open problem; a practical intermediate step is the tail-safe mixture proposal $q_\lambda = (1-\lambda)q_F + \lambda r$ with a heavy-tailed safety component r, which would provide a full-chain minorisation bound at the cost of slightly reduced acceptance. On bounded parameter spaces (such as the structural identification example with uniform priors), a uniform safety component r yields a trivial full-chain bound, suggesting that the core-to-full-support gap is bridgeable in practice.

Data availability

The synthetic target distributions used in this study are fully specified in Appendix A and can be regenerated deterministically from the equations provided. Code to reproduce all synthetic experiments will be made available in a public repository. The real accelerometer data used in Section 6 are subject to confidentiality restrictions imposed by the building owner and are not publicly available.

Disclosure of AI use

The author used Claude (Anthropic) for language polishing.

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